

Fine Tuning Wazuh

A. Descriptiom

1. Buat Ruleset & Decoder baru untuk 2 log file dibawah ini, dengan konsep atomic ruleset dan siblings decoder
2. Testing dengan logtest atau wazuh log testing di dashboard wazuh.

B. ruleset

- [./decoders/0585-openstack-nova.xml](#)

```
M README.M M > 058-openstack-nova.xml X <> 1001-openstack-nova-rules.xml
```

```
You, 22 hours ago | 1 author (Nov)
```

```
1 <!-- LOG Parent -->
2 <decoder name="nova-api-log">
3   <prematch type="pcr2">*nova-api\.log/<prematch>
4 </decoder>
5
6 <decoder name="nova-computed-log">
7   <prematch type="pcr2">*nova-compute\.log/<prematch>
8 </decoder>
9
10 <!-- A. Decoder nova-api -->
11 <!-- 1. LOG HTTP -->
12 <decoder name="nova-api-log-http">
13   <parent>nova-api-log</parent>
14   <prematch type="pcr2">{"GET|POST|PUT|DELETE"}.* HTTP/[0-9\.\+]*</prematch>
15   <regex type="pcr2">\.([0-9]{4})-[0-9]{2}-[0-9]{2}-[0-9]{2}:[0-9]{2}:([0-9]{2}):([0-9]{2}):([0-9]{2}) ([@-9]{4,6}) (INFO|DEBUG|ERROR|WARNING|CRITICAL) (([\w\+]+) \[req-([a-f0-9\-\+])?(:? [\^\]]*)?)?: ([^\]]*)?(?: \|instance: ([a-f0-9\-\+])\}))?(?: (.+)))/</regex>
16   <order>log_timestamp, event_timestamp, pid, log_level, module, request_id, ip, http_method, http_path, http_version, http_status, http_len, http_time</order>
17 </decoders>
18
19 <!-- 99. LOG Detail -->
20 <decoder name="nova-api-log-detail">
21   <parent>nova-api-log</parent>
22   <regex type="pcr2">\.(\d{4})-(\d{2})-(\d{2})_(\d{2}):(\d{2}):(\d{4})-(\d{2})-(\d{2})-(\d{2}):(\d{2}):(\d{3}) (\d+) (INFO|DEBUG|ERROR|WARNING|CRITICAL) (([\w\+]+)?:(? \|req-([a-f0-9\-\+])?(:? [\^\]]*)?)?(?: \|instance: ([a-f0-9\-\+])\}))?(?: (.+)))/</regex>
23   <order>log_timestamp, event_timestamp, pid, log_level, module, request_id, instance_id, message</order>
24 </decoder>
25
26 <!-- B. Decoder nova-compute -->
27 <!-- 99. LOG Detail -->
28 <decoder name="nova-compute-log-detail">
29   <parent>nova-computed-log</parent>
30   <regex type="pcr2">\.(\d{4})-(\d{2})-(\d{2})_(\d{2}):(\d{2}):(\d{4})-(\d{2})-(\d{2})-(\d{2}):(\d{2}):(\d{3}) (\d+) (INFO|DEBUG|ERROR|WARNING|CRITICAL) (([\w\+]+)?:(? \|req-([a-f0-9\-\+])?(:? [\^\]]*)?)?(?: \|instance: ([a-f0-9\-\+])\}))?(?: (.+)))/</regex>
31   <order>log_timestamp, event_timestamp, pid, log_level, module, request_id, instance_id, message</order>
32 </decoder>
```

- [./ruleset/1001-openstack-nova-rules.xml](#)

```
M1 README.m1 0585-opentack-nova.xml 1001-opentack-nova-rules.xml
```

```
You, 10 hours ago | 1 author (You)
1 <group name="nova,">
2 <!-- A. Decode As LVL-0 -->
3 <!-- 1. nova-api -->
4 <rule id="100100" level="0">
5   <decoded_as>nova-api-log</decoded_as>
6   <description>openStack Nova API base log</description>
7 </rule>
8
9 <!-- 2. nova-compute -->
10 <rule id="100200" level="0">
11   <decoded_as>nova-computed-log</decoded_as>
12   <description>openStack Nova Compute base log</description>
13 </rule>
14
You, 10 hours ago • update yara
15 <!-- B. Alert - LOG_LEVEL LVL-1,5,8 -->
16 <!-- 1. nova-api -->
17 <rule id="100101" level="1">
18   <if_sid>100100</if_sid>
19   <field name="log_level">INFO</field>
20   <description>nova Compute INFO log</description>
21 </rule>
22
23 <rule id="100102" level="5">
24   <if_sid>100100</if_sid>
25   <field name="log_level">WARNING</field>
26   <description>nova API WARNING log</description>
27 </rule>
28
29 <rule id="100103" level="8">
30   <if_sid>100100</if_sid>
31   <field name="log_level">ERROR</field>
32   <description>nova API ERROR log</description>
33 </rule>
34
35 <!-- 2. nova-compute -->
36 <rule id="100201" level="1">
37   <if_sid>100200</if_sid>
38   <field name="log_level">INFO</field>
39   <description>nova Compute INFO log</description>
40 </rule>
41
42 <rule id="100202" level="5">
43   <if_sid>100200</if_sid>
44   <field name="log_level">WARNING</field>
45   <description>nova Compute WARNING log</description>
46 </rule>
47
48 <rule id="100203" level="8">
49   <if_sid>100200</if_sid>
50   <field name="log_level">ERROR</field>
51   <description>nova Compute ERROR log</description>
52 </rule>
53 </group>
```

C. how to setup

1. setup & install wazuh-manager

a. wazuh manager only with Docker

- buat container terlebih dahulu dengan ketentuan seperti ini: image=wazuh/wazuh-manager, volume=wazuh:/var/ossec (opsional), dan port forwardnya seperti dibawah
- lalu setelah terbuat jangan lupa install nano karena belum ada nano

```
mkdir -p wazuh
docker run -d --name wazuh-manager \
  -p 1514:1514/udp \
  -p 1515:1515 \
  -v wazuh:/var/ossec \
  wazuh/wazuh-manager:4.7.3

docker exec -it wazuh-manager apt install nano

### masuk ke shell
docker exec -it wazuh-manager bash
```

b. wazuh with ova (Virtual Box)

- install ova dari url <https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html>
- setelah itu **import di virtual box** dan jalankan
- lalu **sesuaikan network adaptor**
- jangan lupa gunakan **RAM minimal 6Gb / 8Gb, CPU 4, dan Storage 30-50Gb**
- lalu jalankan vm nya
 - kredensial **untuk ssh / login** adalah **wazuh-user:wazuh**
 - kredensial **untuk web** adalah **admin:admin**

2. how to make decoder rules

- pertama tama kita coba buat sebuah rules untuk mendapatkan string awal atau nama program lognya terlebih dahulu dengan regex, saya mencari regexnya dengan bantuan AI. ini adalah regex yang saya dapatkan `^nova\[^\ ]+\.\log`

- jika sudah coba lakukan di [regex101](#) dan test dengan salah satu log nova-api



- jika sudah dapet kita coba buat rules decoder parent nya dulu

```
<decoder name="nova-log">
  <prematch type="pcre2">^nova\[^\s\]+\.\log</prematch>
</decoder>
```

- lalu saya coba simpan di `/var/ossec/ruleset/decoders/0585-openstack-nova.xml`

```
nano /var/ossec/ruleset/decoders/0585-openstack-nova.xml # isi rulesnya
chown -R root:wazuh /var/ossec/ruleset/decoders/0585-openstack-nova.xml #
permission
```

- jika sudah di simpan, dan mengubah permissionya, kita bisa lakukan test dengan menggunakan **logtest cli** dengan perintah

```
echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 00:00:00.008 25746 INFO  
nova.osapi_compute.wsgi.server [req-38101a0b-2096-447d-96ea-a692162415ae  
113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -]  
10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail  
HTTP/1.1" status: 200 len: 1893 time: 0.2477829' | /var/ossec/bin/wazuh-  
logtest
```

```

[root@wazuh-server wazuh-user]# cat << 'EOF' > /var/ossec/ruleset/decoders/0585-openstack-nova.xml
> <decoder name="nova-log">
  <prematch type="pcre2">^nova\[^\ ]+\.\log</prematch>
</decoder>
> EOF
[root@wazuh-server wazuh-user]# chown -R root:wazuh /var/ossec/ruleset/decoders/0585-openstack-nova.xml
[root@wazuh-server wazuh-user]# echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 00:00:00.008 25746 INFO nova.osapi_compute
b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -] 10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers
Starting wazuh-logtest v4.12.0
Type one log per line

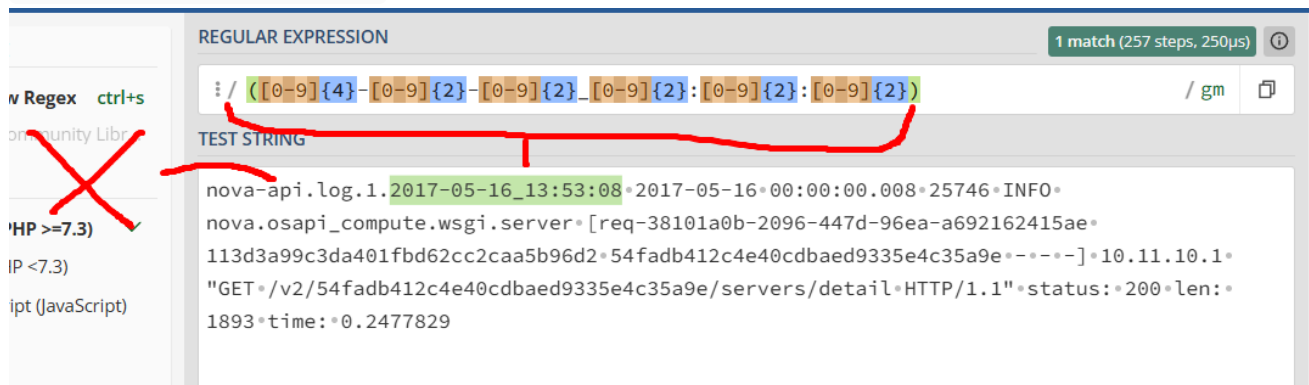
**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 00:00:00.008 25746 INFO nova.osapi_comp
e40cdbaed9335e4c35a9e - - -] 10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail HTTP/1.1"

**Phase 2: Completed decoding.
name: 'nova-log'

[root@wazuh-server wazuh-user]# |

```

- jika sudah kita bisa buat regex untuk dapetin log untuk dapetin data selanjutnya
- kita coba di regex101 lagi dengan regex seperti ini `([0-9]{4}-[0-9]{2}-[0-9]{2}_[0-9]{2}:[0-9]{2}:[0-9]{2})`



- *teks yang **nova-api.log** tidak diambil karena **udh di ambil di name prematch sebelumnya**, dan kita hanya melanjutkan saja bagian selanjutnya
- kita coba dengan menambahkan rule decoder baru seperti ini

```

<decoder name="nova-log-api">
  <parent>nova-log</parent>
  <regex type="pcre2" offset="after_parent">([0-9]{4}-[0-9]{2}-[0-9]{2}_[0-9]{2}:[0-9]{2}:[0-9]{2})</regex>
  <order>log_timestamp</order>
</decoder>

```

- ```
[root@wazuh-server wazuh-user]# cat << 'EOF' >> /var/ossec/ruleset/decoders/0585-openstack-nova.xml
<decoder name="nova-log-api">
 <parent>nova-log</parent>
 <regex type="pcre2" offset="after_parent">([0-9]{4}-[0-9]{2}-[0-9]{2})_([0-9]{2}:([0-9]{2}:([0-9]{2}))</regex>
 <order>log_timestamp</order>
</decoder>
EOF
[root@wazuh-server wazuh-user]# echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 00:00:00.008 25746 INFO nova
b96d2 54fadb412c4e40cdbaed9335e4c35a9e - -] 10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/deta
Starting wazuh-logtest v4.12.0
Type one log per line

**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 00:00:00.008 25746 INFO nova.osapi_compute.w
e40cdbaed9335e4c35a9e - -] 10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail HTTP/1.1" stat

**Phase 2: Completed decoding.
name: 'nova-log'
log_timestamp: '2017-05-16_13:53:08'

[root@wazuh-server wazuh-user]#
```

- ```
REGULAR EXPRESSION
1 match (124 steps, 565µs)

: / \.([0-9]{4}-[0-9]{2}-[0-9]{2}-[0-9]{2}):[0-9]{2}:([0-9]{2})\.([0-9]{4}-[0-9]{2}-[0-9]{2}):[0-9]{2}:([0-9]{2})\.([0-9]{3})\.([0-9]{4,6})
4-([INFO|DEBUG|ERROR|WARNING|CRITICAL])\.([w\.\.]+)\.([req-([a-f0-9-]+)])(?:
([^\]]*)?)\.(?:([0-9\.\.]+))?"(GET|POST|PUT|DELETE)\.([^\.]+)\.HTTP\[([0-9.]
9.]+)"*status:([0-9]{3})*len:([0-9]{1,5})*time:([0-9\.\.]+)
```

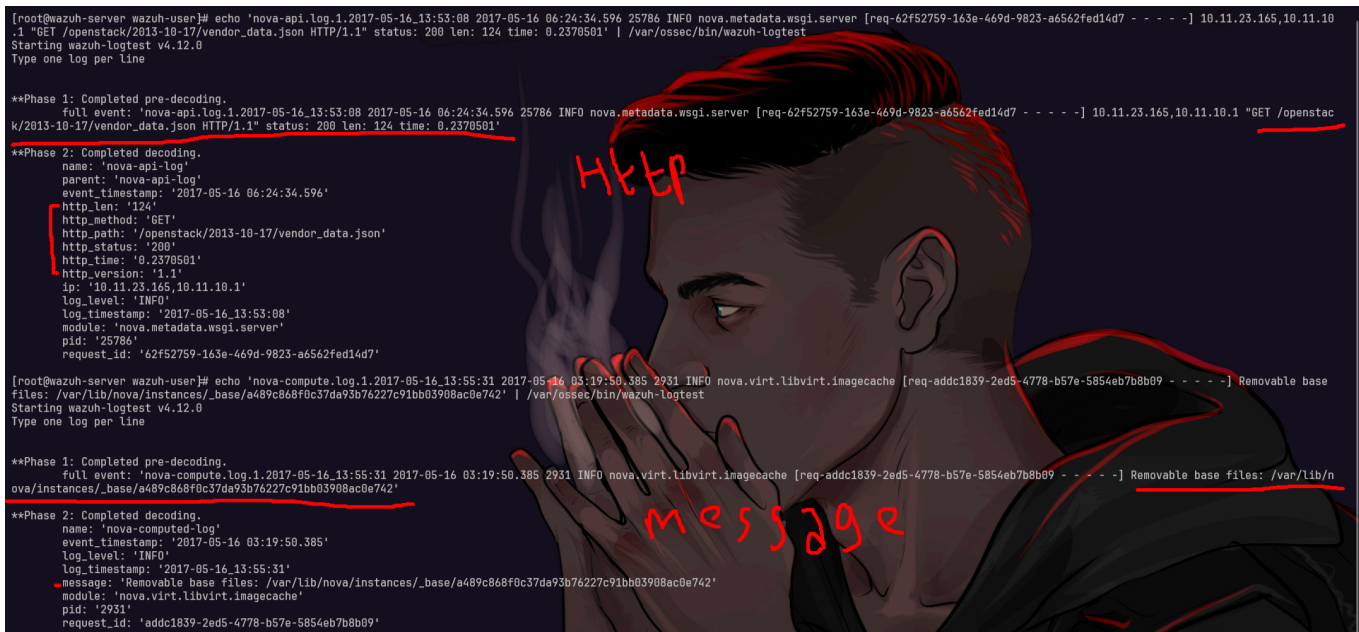
TEST STRING

```
nova-api.log.1.2017-05-16_13:53:08*2017-05-16*00:00:00.008*25746*INFO*
nova.osapi_compute.wsgi.server*[req-38101a0b-2096-447d-96ea-a692162415ae*
113d3a99c3da401fbd62cc2caa5b96d2*54fadb412c4e40cdbaed9335e4c35a9e*---]*10.11.10.1*
"GET*/v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail*HTTP/1.1*status:200*len:*
1893*time:*0.2477829
```

file rulesnya ada di link diatas

lalu lakukan **test** dengan beberapa log yang berbeda karena ada log dari **nova-api**, dan **nova-compute**

```
echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO
nova.metadata.wsgi.server [req-62f52759-163e-469d-9823-a6562fed14d7 - - - -]
10.11.23.165,10.11.10.1 "GET /openstack/2013-10-17/vendor_data.json HTTP/1.1"
status: 200 len: 124 time: 0.2370501' | /var/ossec/bin/wazuh-logtest
echo 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:50.385 2931 INFO
nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09 - - - -]
Removable base files:
/var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742' |
/var/ossec/bin/wazuh-logtest
```



```
[root@wazuh-server wazuh-user]# echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO nova.metadata.wsgi.server [req-62f52759-163e-469d-9823-a6562fed14d7 - - - -] 10.11.23.165,10.11.10.1 "GET /openstack/2013-10-17/vendor_data.json HTTP/1.1" status: 200 len: 124 time: 0.2370501' | /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.12.0
Type one log per line

**Phase 1: Completed pre-decoding.
  full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO nova.metadata.wsgi.server [req-62f52759-163e-469d-9823-a6562fed14d7 - - - -] 10.11.23.165,10.11.10.1 "GET /openstack/2013-10-17/vendor_data.json HTTP/1.1" status: 200 len: 124 time: 0.2370501'
**Phase 2: Completed decoding.
  name: 'nova-api-log'
  parent: 'nova-api-log'
  event_timestamp: '2017-05-16 06:24:34.596'
  http_len: '124'
  http_method: 'GET'
  http_path: '/openstack/2013-10-17/vendor_data.json'
  http_status: '200'
  http_time: '0.2370501'
  http_version: '1.1'
  ip: '10.11.23.165,10.11.10.1'
  log_level: 'INFO'
  log_timestamp: '2017-05-16 13:53:08'
  module: 'nova.metadata.wsgi.server'
  pid: '25786'
  request_id: '62f52759-163e-469d-9823-a6562fed14d7'

[root@wazuh-server wazuh-user]# echo 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:50.385 2931 INFO nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09 - - - -] Removable base files: /var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742' | /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.12.0
Type one log per line

**Phase 1: Completed pre-decoding.
  full event: 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:50.385 2931 INFO nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09 - - - -] Removable base files: /var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742'
**Phase 2: Completed decoding.
  name: 'nova-compute-log'
  event_timestamp: '2017-05-16 03:19:50.385'
  log_level: 'INFO'
  log_timestamp: '2017-05-16 13:55:31'
  message: 'Removable base files: /var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742'
  module: 'nova.virt.libvirt.imagecache'
  pid: '2931'
  request_id: 'addc1839-2ed5-4778-b57e-5854eb7b8b09'
```

di gambar di atas terdapat 2 log yaitu nova-api, dan nova-compute yang **nova api** itu mengarah ke **HTTP**, sedangkan nova-compute lebih ke **message**

4. how to make rulesets

- pertama tama kita test dengan membaut ruleset yang mendeteksi **LOG_LEVEL=INFO**, untuk rulanya kurang lebih seperti ini

```
<group name="nova,">
  <!-- Untuk nova-api -->
  <rule id="100100" level="0">
    <decoded_as>nova-api-log</decoded_as>
    <description>OpenStack Nova API base log</description>
  </rule>
```

```

<!-- Untuk nova-compute -->
<rule id="100200" level="0">
  <decoded_as>nova-computed-log</decoded_as>
  <description>OpenStack Nova Compute base log</description>
</rule>

<rule id="100101" level="1">
  <if_sid>100100</if_sid>
  <field name="log_level">INFO</field>
  <description>Nova Api INFO log</description>
</rule>

<rule id="100201" level="1">
  <if_sid>100200</if_sid>
  <field name="log_level">INFO</field>
  <description>Nova Compute INFO log</description>
</rule>
</group>

```

- disini kita buat filenya dulu dan ubah permissionya

```

nano /var/ossec/ruleset/rules/1001-openstack-nova-rules.xml
chown -R root:wazuh /var/ossec/ruleset/rules/1001-openstack-nova-rules.xml

# atau gunakan output redirection
cat << 'EOF' > /var/ossec/ruleset/rules/1001-openstack-nova-rules.xml
## copy paste rulesnya
EOF

```

- lalu lakukan test dengan menggunakan **wazuh-logtest** untuk memastikan bahwa log yang kita buat sebelum

- kita bisa gunakan perintah yang sebelumnya

```
[root@wazuh-server wazuh-user]# cat << 'EOF' > /var/ossec/ruleset/rules/1001-openstack-nova-rules.xml
> <group name="nova,">
  <!-- Untuk nova-api -->
  <rule id="100100" level="0">
    <decoded_as>nova-api-log</decoded_as>
    <description>OpenStack Nova API base log</description>
  </rule>

  <!-- Untuk nova-compute -->
  <rule id="100200" level="0">
    <decoded_as>nova-computed-log</decoded_as>
    <description>OpenStack Nova Compute base log</description>
  </rule>

  <rule id="100101" level="1">
    <if_sid>100100</if_sid>
    <field name="log_level">INFO</field>
    <description>Nova Api INFO log</description>
  </rule>

  <rule id="100201" level="1">
    <if_sid>100200</if_sid>
    <field name="log_level">INFO</field>
    <description>Nova Compute INFO log</description>
  </rule>
</group>
> EOF
[root@wazuh-server wazuh-user]# echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO
.1 "GET /openstack/2013-10-17/vendor_data.json HTTP/1.1" status: 200 len: 124 time: 0.2370501' | /var/ossec
Starting wazuh-logtest v4.12.0
Type one log per line

**Phase 1: Completed pre-decoding.
  full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO nova.metadata.wsgi.server
k/2013-10-17/vendor_data.json HTTP/1.1" status: 200 len: 124 time: 0.2370501'

**Phase 2: Completed decoding.
  name: 'nova-api-log'
  parent: 'nova-api-log'
  event_timestamp: '2017-05-16 06:24:34.596'
  http_len: '124'
  http_method: 'GET'
  http_path: '/openstack/2013-10-17/vendor_data.json'
  http_status: '200'
  http_time: '0.2370501'
  http_version: '1.1'
  ip: '10.11.23.165,10.11.10.1'
  log_level: 'INFO'
  log_timestamp: '2017-05-16_13:53:08'
  module: 'nova.metadata.wsgi.server'
  pid: '25786'
  request_id: '62f52759-163e-469d-9823-a6562fed14d7'

**Phase 3: Completed filtering (rules).
  id: '100101'
  level: '1'
  description: 'Nova Api INFO log'
  groups: '['nova']'
  fireddtimes: '1'
```

5. lakukan test dengan rulesets yang sudah dibuat

copy paste rules nya dengan nano, atau output redirection, samain aja sama perintah di atas
trus copy paste rulesnya

file rulesnya ada di link diatas

lalu lakukan **test dengan beberapa log yang berbeda beda** jenisnya saya sudah mengumpulkan **beberapa log yang berbeda** seperti ada **INFO, WARNING< ERROR, dan HTTP**

```
# INFO
## normal
echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO
nova.metadata.wsgi.server [req-62f52759-163e-469d-9823-a6562fed14d7 - - - - -]
10.11.23.165,10.11.10.1 "GET /openstack/2013-10-17/vendor_data.json HTTP/1.1"
status: 200 len: 124 time: 0.2370501' | /var/ossec/bin/wazuh-logtest
echo 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:50.385 2931 INFO
nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09 - - - - -]
Removable base files:
/var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742' |
/var/ossec/bin/wazuh-logtest

## abnormal
echo 'nova-api.log.2017-05-14_21:27:04 2017-05-14 19:39:01.445 25746 INFO
nova.osapi_compute.wsgi.server [req-5a2050e7-b381-4ae9-92d2-8b08e9f9f4c0
113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -]
10.11.10.1 "GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail HTTP/1.1"
status: 200 len: 1583 time: 0.1919448' | /var/ossec/bin/wazuh-logtest
echo 'nova-compute.log.2017-05-14_21:27:09 2017-05-14 19:39:02.007 2931 INFO
nova.virt.libvirt.driver [req-e285b551-587f-4c1d-8eba-dceb2673637f
113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -]
[instance: 3edec1e4-9678-4a3a-a21b-a145a4ee5e61] Creating image' |
/var/ossec/bin/wazuh-logtest

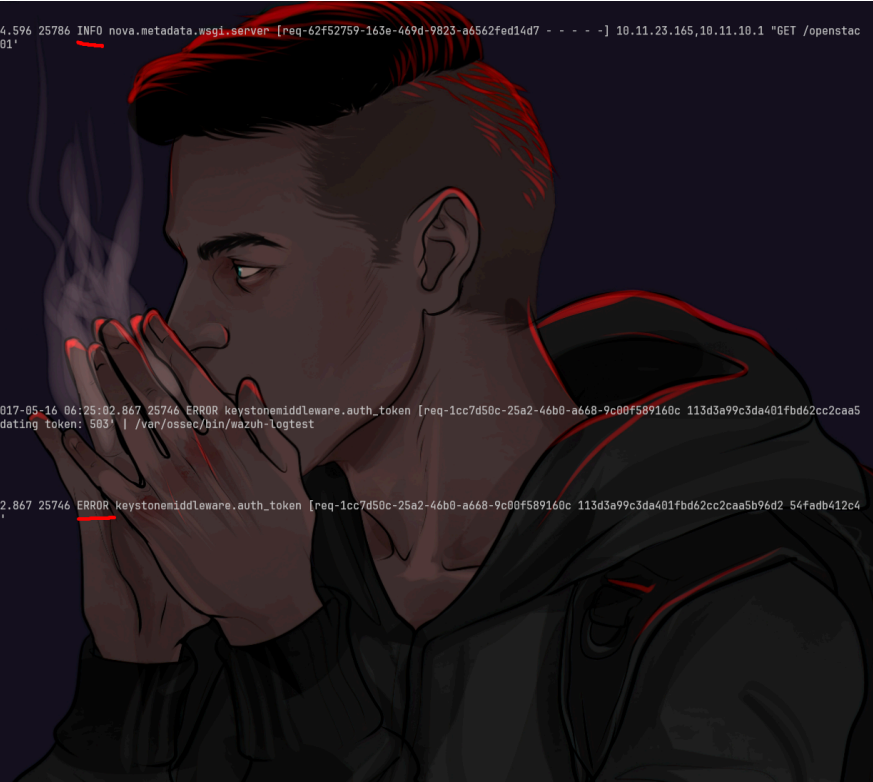
# ERROR
## normal
echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:25:02.867 25746 ERROR
keystone.middleware.auth_token [req-1cc7d50c-25a2-46b0-a668-9c00f589160c
113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -] Bad
response code while validating token: 503' | /var/ossec/bin/wazuh-logtest
echo 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:45.356 2931
ERROR oslo_service.periodic_task Traceback (most recent call last):' |
/var/ossec/bin/wazuh-logtest

# warning
## normal
echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:25:02.868 25746 WARNING
keystone.middleware.auth_token [req-1cc7d50c-25a2-46b0-a668-9c00f589160c
113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4e40cdbaed9335e4c35a9e - - -]
Identity response: <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">' |
/var/ossec/bin/wazuh-logtest
echo 'nova-compute.log.1.2017-05-16_13:55:31 2017-05-16 03:19:50.384 2931
```

```
WARNING nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09
- - - - -] Unknown base file:
/var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742' |
/var/ossec/bin/wazuh-logtest
```

```
## abnormal
```

```
echo 'nova-compute.log.2017-05-14_21:27:09 2017-05-14 19:59:40.806 2931
WARNING nova.virt.libvirt.imagecache [req-addc1839-2ed5-4778-b57e-5854eb7b8b09
- - - - -] Unknown base file:
/var/lib/nova/instances/_base/a489c868f0c37da93b76227c91bb03908ac0e742' |
/var/ossec/bin/wazuh-logtest
```



```
**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:24:34.596 25786 INFO nova.metadata.wsgi.server [req-62f52759-163e-469d-9823-a6562fed14d7 - - - -] 10.11.23.165,10.11.10.1 "GET /openstac
k/2013-10-17/vendor_data.json HTTP/1.1" status: 200 len: 124 time: 0.2370501'

**Phase 2: Completed decoding.
name: 'nova-api-log'
parent: 'nova-api-log'
event_timestamp: '2017-05-16 06:24:34.596'
http_len: '124'
http_method: 'GET'
http_path: '/openstack/2013-10-17/vendor_data.json'
http_status: '200'
http_time: '0.2370501'
http_version: '1.1'
ip: '10.11.23.165,10.11.10.1'
log_level: 'INFO'
log_timestamp: '2017-05-16_13:53:08'
module: 'nova.metadata.wsgi.server'
pid: '25786'
request_id: '62f52759-163e-469d-9823-a6562fed14d7'

**Phase 3: Completed filtering (rules).
id: '100101'
level: '1'
description: 'Nova Compute INFO log'
groups: '["nova"]'
firedtimes: '1'
mail: 'False'

[root@wazuh-server wazuh-user]# echo 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:25:02.867 25746 ERROR keystone.middleware.auth_token [req-1cc7d50c-25a2-46b0-a668-9c00f589160c 113d3a99c3da401fbd62cc2caa5
b96d2 54fadb412c4e40cdcbad9335e4c35a9e - - - -] Bad response code while validating token: 503' | /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.12.0
Type one log per line

**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.1.2017-05-16_13:53:08 2017-05-16 06:25:02.867 25746 ERROR keystone.middleware.auth_token [req-1cc7d50c-25a2-46b0-a668-9c00f589160c 113d3a99c3da401fbd62cc2caa5b96d2 54fadb412c4
e40cdcbad9335e4c35a9e - - - -] Bad response code while validating token: 503'

**Phase 2: Completed decoding.
name: 'nova-api-log'
event_timestamp: '2017-05-16 06:25:02.867'
log_level: 'ERROR'
log_timestamp: '2017-05-16_13:53:08'
message: 'Bad response code while validating token: 503'
module: 'keystone.middleware.auth_token'
pid: '25746'
request_id: '1cc7d50c-25a2-46b0-a668-9c00f589160c'

**Phase 3: Completed filtering (rules).
id: '100103'
level: '8'
description: 'Nova API ERROR log'
groups: '["nova"]'
firedtimes: '1'
mail: 'False'

**Alert to be generated.
```

bisa juga gunakan di webnya ada di [<ip_wazuh>/app/ruleset-test#/wazuh-dev?tab=logtest](#)

```
nova-api.log-2017-05-14_21:27:04 2017-05-14 19:59:26.844 25746 INFO nova.osapi.compute.wsgi.server [req-1cda9e00-065c-4f15-b673-f22cd3c3bc33 1133da99c3da4fd16d2cc2caa5b96d2 54fab4d42c4a4dcdbae9335e4c35a9e - -] 10.11.0.11 "GET /v2/54fab4d42c4a4dcdbae9335e4c35a9e/servers/detail/1?status=200 len: 1893 time: 0.2585850"
nova-api.log-2017-05-14_21:27:04 2017-05-14 19:59:27.961 25743 INFO nova.openstack.compute.server_external_events [req-d6ef861e-9d25-426d-902c-dc7396face61 17b8d1f1d444643b07a10ca70d2f1e9f746793ac574c6b8a9e857f56a7608 - -]
Creating event network-vl-plugged:7df8c35e-1f99-4102-b86e-846de695c2c6 for instance 45953531-5fe1-48ed-42e2-5577ae4a22b0
nova-api.log-2017-05-14_21:27:04 2017-05-14 19:59:27.962 25743 INFO nova.openstack.compute.server_external_events [req-d6ef861e-9d25-426d-902c-dc7396face61 17b8d1f1d444643b07a10ca70d2f1e9f746793ac574c6b8a9e857f56a7608 - -] 10.11.0.11 "POST
```

 Clear session

```

Messages:
INFO: (7282): Session initialized with token 'ef80c3f3'

**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.2017-05-14,21:27:04 2017-05-14 19:59:26.844 25746 INFO nova.osapi_compute.wsgi.server [req-1cdaa9d0-b65c-4f15-a673-f22cd3dc33 113d3a99c3da01fb62cc2caa5096d2 54fadb412c4e48c0baed9335e4c35a9e - - - 10.11.10.1] GET /v2/54fadb412c4e48c0baed9335e4c35a9e/servers/detail HTTP/1.1' status: 200 len: 1093 time: 0.258595'

**Phase 2: Completed decoding.
name: 'nova-api-log'
parent: 'nova-api-log'
event_timestamp: '2017-05-14 19:59:26.844'
http_len: '1093'
http_method: 'GET'
http_path: '/v2/54fadb412c4e48c0baed9335e4c35a9e/servers/detail'
http_status: '200'
http_time: '0.258595'
http_version: '1.1'
ip: '10.11.10.1'
log_level: 'INFO'
log_timestamp: '2017-05-14,21:27:04'
module: 'nova.osapi_compute.wsgi.server'
pid: '25746'
request_id: '1cdaa9d0-b65c-4f15-a673-f22cd3dc33'

**Phase 3: Completed filtering (rules).
id: '100101'
level: '1'
description: 'Nova Compute INFO log'
groups: ['nova']
firedtimes: '1'
mail: 'false'

**Phase 1: Completed pre-decoding.
full event: 'nova-api.log.2017-05-14,21:27:04 2017-05-14 19:59:27.961 25743 INFO nova.api.openstack.compute.server_external_events [req-ed6fe816-9d25-426d-802c-de7396bf6ace 67b8d1fd44d44643b07fa18ca7d821fb e9746973ac52c74c6bb9e8857f56a7088 - - -] Creating event network-vif-plugged:7df8c35e-f199-4f02-b86e-884de6f95c26 for instance 4d59535f-5fef-48ed-2ae-557fe4e22b0'

**Phase 2: Completed decoding.
name: 'nova-api-log'

```